

Specular Glint–Constrained Attitude Inversion

Weekly Report — 20 March 2026

Girish Narayanan

Summary

All experiments now use **realistic trajectories** (micro46 dataset, $|\omega| \in [0.1, 1.5]$ deg/s, 100 random attitudes). The main result this week: a concrete pipeline architecture for attitude inversion anchored on specular glint physics.

Specular Glint Physics

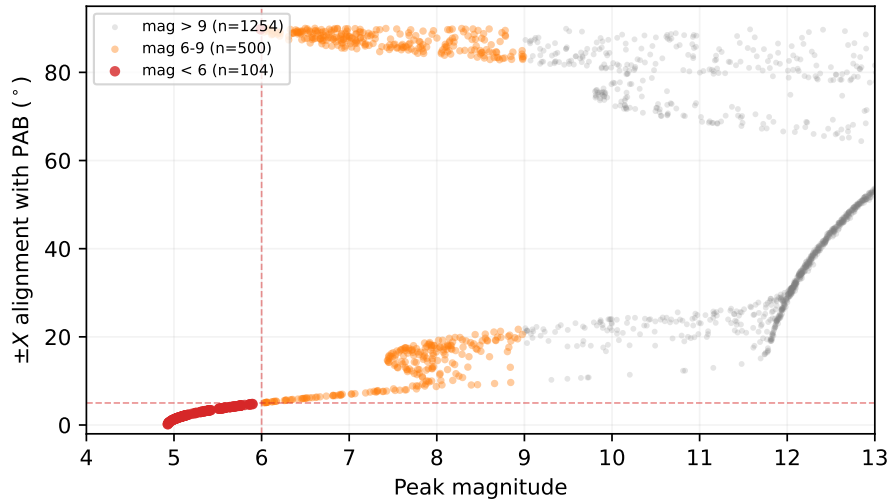


Figure 1: $\pm X$ alignment with PAB at all 1858 brightness peaks across 100 trajectories. Every peak brighter than mag 6 has $\pm X$ within 5° of the PAB (104/104, zero exceptions).

Key facts (verified across 100 trajectories, 104 specular peaks):

- $\text{mag} < 6 \Rightarrow \pm X$ aligned within 5° of PAB (100%)
- Always the $\pm X$ bus faces (99%; one $+Z$ exception)
- Attitude constrained to a 1-DOF circle at each specular epoch
- 62/100 trajectories have ≥ 1 specular glint

Pipeline Architecture

Step 1 — Peak detection and $|\omega|$ estimation

- Detect brightness peaks (local minima, prominence > 0.3 mag)
- Count total peaks n_{peaks}
- $|\omega|_{\text{est}} = 0.040 \times n_{\text{peaks}} + 0.042$ deg/s (calibrated on 100 trajectories, $\rho = 0.92$, median error 13%)
- Define magnitude search band: $|\omega|_{\text{est}} \times [0.8, 1.2]$

Step 2 — Anchor selection

- a. Find all peaks with $\text{mag} < 6.0$ (guaranteed $\pm X$ specular, Section 2)
- b. Select brightest as anchor epoch t_a
- c. Constraint: at t_a , the $\pm X$ normal is aligned with PAB $\Rightarrow$ attitude lies on a 1-DOF circle parameterised by twist ϕ

Step 3 — Attitude candidates at anchor (hi-fi pre-filter)

- a. Generate $2 \times 360 = 720$ candidates: $\{+X, -X\} \times \phi \in [0^\circ, 360^\circ)$ at 1° spacing
- b. For each: single-epoch hi-fi brightness eval at t_a (~ 50 ms/eval, 36 s total)
- c. Keep candidates matching observed brightness within factor 2 $\Rightarrow \sim 100$ survivors
- d. Verified: correct attitude within 1.5° of a survivor

Step 4 — ω grid search via glint alignment cost

- a. Grid: N_{dir} Fibonacci directions $\times N_{\text{mag}}$ magnitudes in search band
- b. **For each** ω_{test} :
 - i. Propagate identity quaternion from t_a with ω_{test} (Euler dynamics) $\Rightarrow$ relative rotation Δq at each non-anchor specular glint. *Key: Δq is independent of q_{anchor} — one propagation serves all ϕ candidates.*
 - ii. **For each** surviving ϕ : $q_{\text{glint}} = q_{\text{anchor}}(\phi) \otimes \Delta q$; check $\pm X$ alignment with PAB
 - iii. Record best (ϕ, cost)
- c. Glint cost: $C = \sum_i (1 - \max(\hat{n}_{+X} \cdot \hat{p}_i, \hat{n}_{-X} \cdot \hat{p}_i))^2$
- d. At truth: $C \approx 3 \times 10^{-5}$. At 2° error: $C > 2$ (Section 4)

Step 5 — Nelder–Mead refinement

- a. Take top 50 by glint cost
- b. Optimise 3D ω via Nelder–Mead, minimising glint cost
- c. ~ 50 – 200 propagations each, ~ 2 s per start

Step 6 — LC scoring and $\pm X$ disambiguation

- a. Back-propagate top 20 to $t = 0$ (Euler dynamics)
- b. Score by lo-fi (no shadows) LC residual over 500 epochs $\Rightarrow$ top 5
- c. Score top 5 by hi-fi (ray-traced shadows) $\Rightarrow$ winner
- d. Run Steps 3–6 twice ($+X$ anchor, $-X$ anchor); pick lower hi-fi residual

Step	Time (8 cores)
Hi-fi pre-filter (720 evals)	36 s
Grid ($2000 \times 20 = 40\text{k}$ propagations)	~ 3 min
Nelder–Mead (50 starts)	~ 4 min
Lo-fi + hi-fi scoring	~ 5 min
Per $\pm X$ hypothesis	~ 13 min
Both hypotheses	~ 26 min

Glint Cost Sensitivity

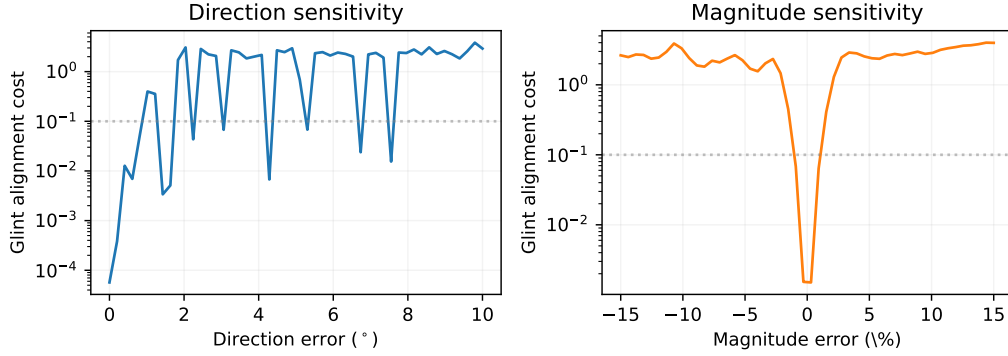


Figure 2: Glint alignment cost vs. ω perturbation (seed 93, 6 non-anchor specular glints). The cost is $\sim 10^{-5}$ at truth and explodes beyond $\sim 2^\circ$ direction or $\sim 1\%$ magnitude error. This extreme sensitivity is both a strength (high discrimination) and a challenge (dense grid required).

The narrow basin requires $N_{\text{dir}} \geq 2000$ ($\sim 2^\circ$ spacing) and $N_{\text{mag}} \geq 20$ ($\sim 2\%$ spacing).

Euler Precession

IS-901 has triaxial asymmetry = 0.556. The body-frame ω direction drifts 56° in 195s at $|\omega| = 1.28 \text{ deg/s}$. The magnitude is approximately conserved. This means the ω recovered at the anchor epoch must be back-propagated via Euler dynamics to obtain ω_0 .

Result (seed 93, +X anchor)

Full pipeline (micro68): grid search (8 cores, 260s) $\rightarrow$ Nelder–Mead $\rightarrow$ phi-sweep $\rightarrow$ lo-fi $\rightarrow$ hi-fi. Total: **7.5 min** (from checkpoint).

Key result: correct ω recovered and present in final hi-fi stage

- Grid found correct ω at rank #13/40k (4.3° , cost 0.029)
- Nelder–Mead refined to cost 3.1×10^{-5} , **ω direction error** = 1.8° , magnitude error 1.3%
- This correct ω candidate (labelled ω_1) **survived all pipeline stages** and reached the final hi-fi scoring round

The full hi-fi results table (top 5 by lo-fi):

Rank	Candidate	Hi-fi	q_0 err	ω err	Note
1	$\omega_2, \phi=300^\circ$	0.383	179.6°	88.1°	Wrong ω , wrong q
2	$\omega_1, \phi=150^\circ$	0.677	88.0°	1.5°	Correct ω
3	$\omega_2, \phi=200^\circ$	1.065	79.7°	88.1°	
4	$\omega_3, \phi=200^\circ$	1.065	79.7°	88.1°	
5	$\omega_3, \phi=10^\circ$	2.330	110.5°	88.1°	

The correct angular velocity is found and survives to the final stage. The pipeline selected the wrong winner only because lo-fi/hi-fi residuals favoured a degenerate candidate (rank 1, hi-fi = 0.383) over the correct one (rank 2, hi-fi = 0.677).

Bottleneck: lo-fi phi selection

The phi-sweep for ω_1 (correct ω) produced 4 candidates:

ϕ	Lo-fi	q_0 err	Note
150°	0.727	88.0°	$\leftarrow$ best lo-fi, wrong q
40°	2.756	22.1°	good q , but poor lo-fi
30°	2.958	32.1°	good q , but poor lo-fi
300°	3.300	122.0°	

The lo-fi residual strongly prefers $\phi = 150^\circ$ (q_0 error 88°) over $\phi = 40^\circ$ (q_0 error 22°). Only the $\phi=150^\circ$ candidate makes it through the lo-fi gate to hi-fi. **The lo-fi scoring stage is the bottleneck:** it discards the correct attitude in favour of an incorrect one that happens to produce a lower no-shadow LC residual.

Implications

1. **The glint-constrained grid search works.** The correct ω is recovered to 1.8° direction / 1.3% magnitude from a blind 40k-point grid in 260 s (8 cores). This is the main result.
2. **The phi-sweep + glint cost works.** The correct ω paired with all phi values have identical glint costs (3.1×10^{-5}), confirming the 1-DOF circle constraint.
3. **Lo-fi LC scoring is an unreliable discriminator for ϕ .** The no-shadow approximation introduces systematic bias that prefers the wrong attitude. Possible fixes: (a) use hi-fi for phi selection (expensive but feasible for ~ 36 candidates), (b) use a short-window hi-fi evaluation, (c) use glint-epoch-only brightness matching instead of full-LC residual.
4. **The final winner selection also needs improvement.** Even among hi-fi candidates, the wrong ω beat the correct one (0.383 vs 0.677). This may be addressable by adding the glint alignment cost as a secondary criterion or penalty term.

Next Steps

1. **Fix phi selection:** replace lo-fi gate with hi-fi short-window or glint-brightness matching to correctly identify the best ϕ for each ω .
2. **Fix final ranking:** incorporate glint cost as tiebreaker or penalty in hi-fi scoring (the correct ω has cost 3×10^{-5} vs 3×10^{-5} for the wrong one — both are degenerate on glint cost alone, but ω direction error could be cross-validated).
3. Test across multiple seeds; establish success rate for ω recovery (the main validated result).
4. Handle trajectories without specular glints (38% of cases).